Composite areas, openings and surface coverage

Day 18 • 2026-11-30 • 2:15 PM–3:30 PM

15 questions. Show your method and include units. For multiple choice, enter A, B, C or D. Follow your instructor’s calculator guidance.

Name _____ **Date** _____

Type into the fields and save a copy, or print and write.

01 READ & REASON

How is the net paintable area of a wall with a window found?

A. Add the window area to the wall area

B. Subtract the window perimeter

C. Divide wall area by window area

D. Subtract the window area from the whole wall area

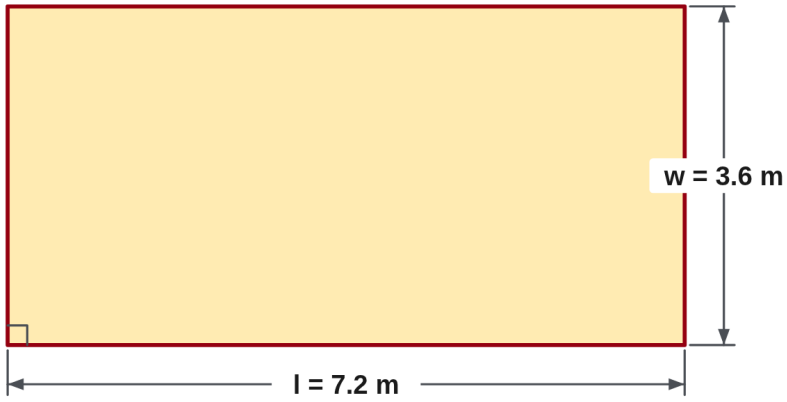
Choice (A–D) _____ **Working / reason** _____

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Composite areas, openings and surface coverage

Day 18 • 2026-11-30 • 15 questions

02 The Shape Canvas floor is 7.2 m by 3.6 m. A 1.2 m by 0.9 m opening is excluded. Find the remaining area.



Shape Canvas

Answer (m²)

Working / reason

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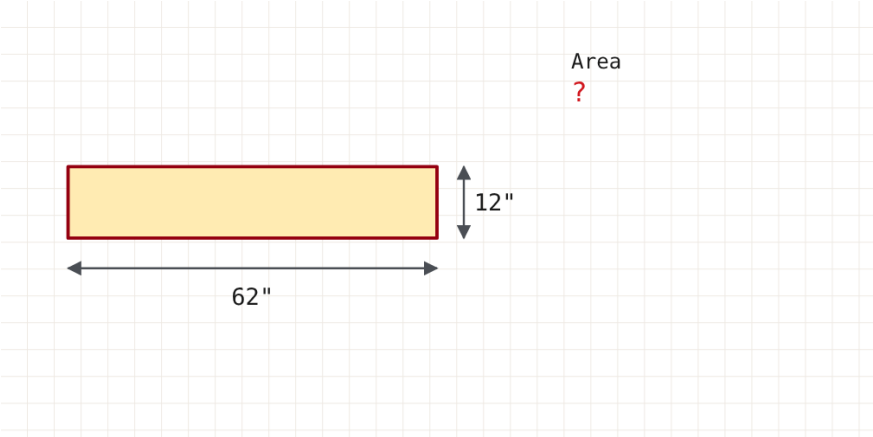
03 The Numeracy Hub area visual supports splitting a composite surface. What must be avoided when adding the parts?

A. Checking with subtraction

B. Counting overlapping area twice

C. Using square units

D. Labelling the dimensions



Numeracy Hub

Choice (A–D)

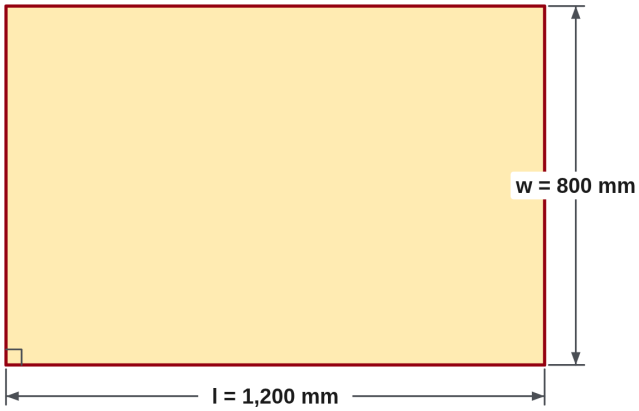
Working / reason

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04 A panel has the Shape Canvas rectangle’s 1200 mm by 800 mm outside dimensions. A 200 mm by 300 mm opening is removed. Find net area in mm².



Shape Canvas

Answer (mm²)

Working / reason

Textbook sections 10.4–10.7 pp.246–254

05 Which units can be subtracted directly from 12 m²?

A. 300 mm

B. 3 m

C. 3 m³

D. 3 m²

Choice (A–D)

Working / reason

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06 CALCULATE

A wall is 5 m by 2.4 m and has a 0.9 m by 2.1 m doorway. Find net wall area.

Answer (m²) **Working / reason**

Textbook sections 10.4–10.7 pp.246–254

07 An L-shaped floor is a 6 m by 5 m rectangle with a 2 m by 3 m corner removed. Find its area.

Answer (m²) **Working / reason**

Textbook sections 10.4–10.7 pp.246–254

08 A gable consists of a 4 m by 2.4 m rectangle topped by a triangle of base 4 m and height 1.2 m. Find total area.

Answer (m²) **Working / reason**

Textbook sections 10.4–10.7 pp.246–254

09 A 7 m by 2.7 m wall contains two 1.2 m by 1.5 m windows. Find net area.

Answer (m²) **Working / reason**

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10 A closed rectangular box is 2 m by 1 m by 0.5 m. Use SA = 2(LW + LH + WH). Find surface area.

Answer (m²)

Working / reason

Textbook sections 10.4–10.7 pp.246–254

11 APPLY & CHECK

A student subtracts a window’s 4 m perimeter from 20 m² of wall area. What is wrong?

A. The subtraction sign must be addition

B. The perimeter needs only rounding

C. The answer must be cubic metres

D. They mixed length with area

Choice (A–D)

Working / reason

Textbook sections 10.4–10.7 pp.246–254

12 A net surface is 18 m². Paint covers 9 m² per litre per coat. Find the theoretical litres for two coats.

Answer (L)

Working / reason

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13 Two rectangles overlap by 2 m². Their separate areas are 12 m² and 9 m². Which expression gives their union area?

A. 12 × 9 ÷ 2

B. 12 + 9 − 2

C. 12 + 9 + 2

D. 12 − 9 − 2

Choice (A–D)

Working / reason

Textbook sections 10.4–10.7 pp.246–254

14 A net cladding area is 22.4 m². Add 10% waste allowance. Find the planned coverage area.

Answer (m²)

Working / reason

Textbook sections 10.4–10.7 pp.246–254

15 A large rectangle minus a cutout and a sum of non-overlapping parts give different areas. What is the best next step?

A. Average the two answers

B. Choose the larger result without checking

C. Ignore the cutout

D. Check dimensions and whether the parts cover exactly the same region

Choice (A–D)

Working / reason

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